

```
# Student Management System
```

```
students = {}
```

```
action_stack = []
```

```
waiting_queue = []
```

```
def add_student():
```

```
    try:
```

```
        roll = int(input("Enter Roll Number: "))
```

```
        name = input("Enter Name: ")
```

```
        mark = float(input("Enter Mark: "))
```

```
        students[roll] = {
```

```
            "name": name,
```

```
            "mark": mark
```

```
        }
```

```
        action_stack.append(f"Added {name}")
```

```
        waiting_queue.append(name)
```

```
        print("Student Added Successfully!")
```

```
    except ValueError:
```

```
        print("Invalid input!")
```

```
def view_students():  
    if not students:  
        print("No students available.")  
        return
```

```
    print("\n--- Student List ---")  
    for roll, data in students.items():  
        print(f"Roll: {roll}")  
        print(f"Name: {data['name']}")  
        print(f"Mark: {data['mark']}")  
        print("-" * 20)
```

```
def search_student():  
    try:  
        roll = int(input("Enter Roll Number: "))  
  
        if roll in students:  
            print(students[roll])  
        else:  
            print("Student not found!")  
  
    except ValueError:  
        print("Invalid Roll Number")
```

```
def delete_student():  
    try:  
        roll = int(input("Enter Roll Number: "))  
  
        if roll in students:  
            name = students[roll]["name"]  
            del students[roll]  
  
            action_stack.append(f"Deleted {name}")  
            print("Student Deleted")  
  
        else:  
            print("Student not found!")  
  
    except ValueError:  
        print("Invalid input")
```

```
def average_mark():  
    if not students:  
        print("No data available")  
        return
```

```
total = 0
```

```
for data in students.values():  
    total += data["mark"]
```

```
avg = total / len(students)
```

```
print("Average Mark =", avg)
```

```
def show_stack():  
    print("\nRecent Actions (Stack)")  
    for item in reversed(action_stack):  
        print(item)
```

```
def show_queue():  
    print("\nWaiting Queue")  
    for item in waiting_queue:  
        print(item)
```

```
while True:  
    print("\n===== STUDENT MANAGEMENT SYSTEM =====")  
    print("1. Add Student")  
    print("2. View Students")  
    print("3. Search Student")  
    print("4. Delete Student")
```

```
print("5. Average Mark")
```

```
print("6. Show Stack")
```

```
print("7. Show Queue")
```

```
print("8. Exit")
```

```
choice = input("Enter Choice: ")
```

```
if choice == "1":
```

```
    add_student()
```

```
elif choice == "2":
```

```
    view_students()
```

```
elif choice == "3":
```

```
    search_student()
```

```
elif choice == "4":
```

```
    delete_student()
```

```
elif choice == "5":
```

```
    average_mark()
```

```
elif choice == "6":
```

```
    show_stack()
```

```
elif choice == "7":
```

```
show_queue()
```

```
elif choice == "8":
```

```
    print("Thank You!")
```

```
    break
```

```
else:
```

```
    print("Invalid Choice")
```